

Neurological and Behavioral Responses to Controlled Financial Volatility and Counterfactual Regret

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Abstract

Traditional discrete neuroeconomic paradigms lack the ecological validity of continuous, volatile financial markets. To bridge this gap, the present study introduces a dynamic Jump-Diffusion Model (JDM) trading task. Ten participants ($N = 10$) navigated sudden simulated market spikes followed by SAM ratings across valence, arousal, and regret to evaluate behavioral responses to increasing monetary stakes (1, 50, 100 DKK), and realized versus counterfactual outcomes (gain, loss, relief, FOMO). Concurrently, real-time 64-channel Electroencephalography (EEG) data were recorded from a single-subject subset ($N = 1$) to evaluate neural responses to the ongoing volatility. Due to the restricted electrophysiological sample, this research serves purely as a methodological pilot study. Behaviorally, the paradigm successfully induced distinct affective states; the counterfactual regret of missing a surge (FOMO) rivaled the distress of a realized loss. However, the *lab money* effect and progressive task habituation meant that increasing monetary stakes failed to modulate arousal ratings. Additionally, prior financial outcomes did not significantly dictate risk behavior, which was instead dominated by an overwhelming action bias. Neurally, the continuous JDM elicited robust late-stage cognitive engagement (P300, LPP), though early prediction error signals (FRN) were masked by ocular tracking artifacts. These preliminary findings validate the JDM as a promising framework for neuroeconomic modeling while establishing critical baselines for fully powered future investigations.

1 Introduction

The gamification of modern retail trading has fundamentally blurred the lines between financial investment and pathological gambling. Traditionally, neuroeconomics has relied on discrete paradigms like the Iowa Gambling Task (IGT) [2]. While successful at demonstrating the neurological basis that guides risk aversion, they fundamentally lack ecological validity when compared to continuous volatile financial markets. Traders do not simply pull a card from a deck; they endure unpredictable market jumps, evaluate counterfactual scenarios in real-time, and experience continuous fluctuations in the psychological valuation of their capital. To bridge this gap, it is necessary to examine how cognitive biases alter subjective emotion as well as the underlying neuronal response in a continuous, unpredictable environment.

1.1 Behavioral Tendencies

Financial decision-making is rarely perfectly rational; it is heavily distorted by prior outcomes and the psychological weight of ownership. Rather than objectively calculating mathematical probabilities, traders navigate continuous market volatility through the lens of three specific cognitive biases.

First, *counterfactual regret* illustrates that the human evaluation of an outcome is strictly relative. Foundational research on regret theory demonstrates that missing out on a lucrative financial surge while holding cash (informally known as the Fear Of Missing Out, or FOMO) generates a counterfactual loss that can rival the psychological distress of an actual realized loss [19, 12]. Conversely, dodging a market crash generates relief, acting as a counterfactual gain.

Second, *the endowment effect* and personal agency demonstrate that individuals consistently ascribe higher value to assets they already own compared to equivalent assets they do not [9]. In a trading environment, having active capital exposure and exercising personal agency (such as actively choosing to buy rather than passively holding) theoretically amplifies the emotional and neurological salience of subsequent price movements.

Third, sequential performance significantly distorts baseline risk calculations. The House Money Effect [17] describes the phenomenon where previous gains act as an emotional buffer, leading to increased risk-seeking behavior in subsequent decisions. Traders perceive themselves to be playing with the market's capital rather than their own. Conversely, prior losses can bidirectionally shift behavior by either inducing acute risk aversion (the Snakebite Effect) or triggering reckless risk-seeking in an attempt to recoup lost capital (the Break-Even Effect).

1.2 Neural Responses to Financial Consequences

These complex financial evaluations manifest in distinct, measurable Event-Related Potentials (ERPs) originating from specific neuroanatomical networks.

Feedback-Related Negativity (FRN): Generated primarily in the Anterior Cingulate Cortex (ACC), the FRN peaks between 200–350 ms post-stimulus and indexes reward prediction errors [7]. The FRN is highly sensitive to negative outcomes and has been shown to process counterfactual losses ([14]), providing a direct metric for comparing the neurological sting of a realized market crash against the regret of a missed surge.

The P300 Complex (P3a/P3b): Occurring 300–600 ms post-stimulus, the P300 component reflects the allocation of attentional resources and is commonly divided into two distinct subcomponents [15]. The fronto-central P3a is driven by stimulus novelty and involuntary attention shifts, which are crucial during sudden volatility spikes. The parietal P3b reflects task-relevance and is hypothesized to scale monotonically with the absolute magnitude or financial stakes of a given reward [18].

Late Positive Potential (LPP): Occurring 400–800 ms post-stimulus, the LPP serves as an index of sustained affective engagement [5]. It captures the prolonged physiological arousal and cognitive processing required when active capital ("skin in the game") is exposed to ongoing market fluctuations.

1.3 The Present Study, Pilot Objective & Hypotheses

To overcome the ecological limitations of discrete economic tasks, this study utilizes a continuous, synthetic Jump-Diffusion Model (JDM) to evaluate neural and behavioral responses to real-time financial volatility. By placing participants in active (invested) and passive (cash) positions during simulated crashes and surges, we aim to separate actual financial loss from counterfactual regret. Financial stakes are escalated across blocks to test outcome scaling.

EEG data collection was restricted to a single participant ($N = 1$) alongside a broader behavioral cohort. Consequently, this functions as a methodological pilot study. The primary objective is to validate the Jump-Diffusion paradigm, that is, proving that simulated continuous markets can successfully elicit time-locked ERPs and to lay the groundwork for fully powered future studies.

Specifically, this study aims to investigate the following hypotheses:

- H1** The continuous Jump-Diffusion model will elicit observable P300 and FRN components, validating its efficacy against established discrete-choice paradigms.
- H2** Favorable outcomes ('Gain' and 'Relief') will yield higher valence, lower regret, and an attenuated FRN, whereas unfavorable outcomes ('Loss' and 'FOMO') will yield lower valence, higher regret, and a more pronounced FRN. Furthermore, active agency is expected to amplify this neural sensitivity, further polarizing the FRN amplitude according to the realized outcome.
- H3** Driven by the endowment effect, active capital exposure (being invested) will amplify overall task salience, yielding higher physiological arousal ratings and larger LPP amplitudes compared to passive observation.
- H4** Increased financial stakes will monotonically increase parietal P3b amplitudes and linearly scale subjective emotional ratings.
- H5** Prior financial outcomes will bidirectionally modulate subsequent risk preferences, with recent realized gains increasing risk-seeking behavior and recent losses systematically distorting baseline risk aversion.

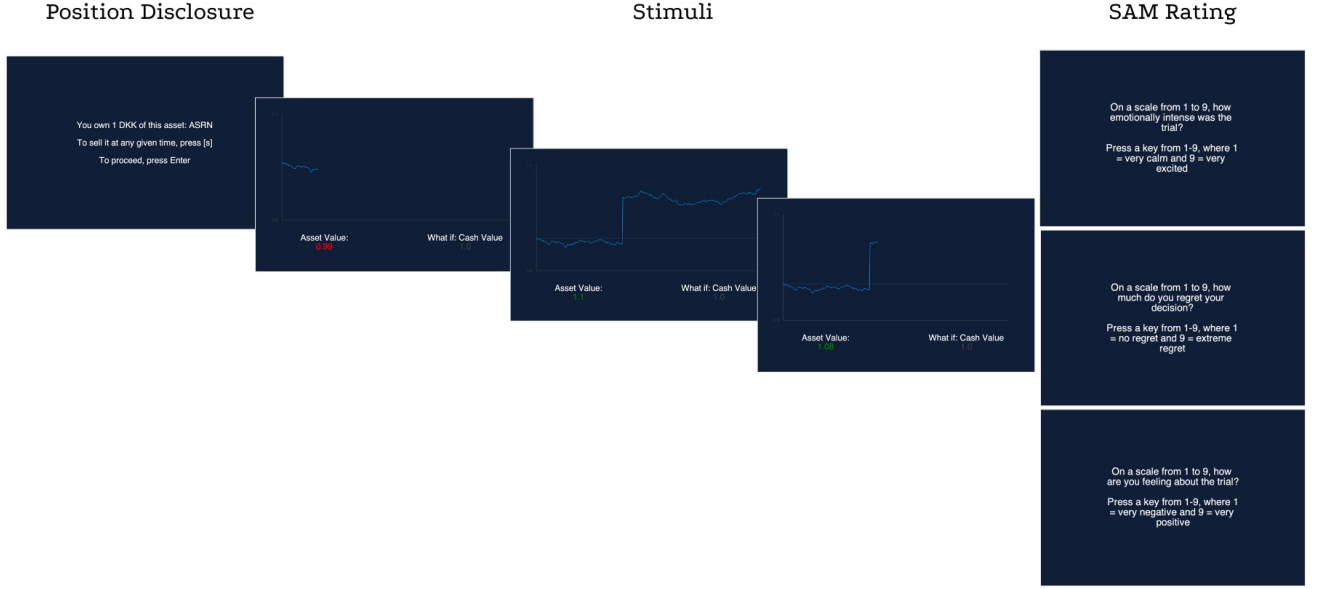


Figure 1: Experimental trial flow. Each trial begins with a **(left)** Position Disclosure screen displaying the fictional asset, player position, and available action. Once confirmed, the **(middle)** Stimuli phase unfolds as a dynamic graph displaying a predetermined asset price spike. Each trial concludes with the **(right)** SAM Rating phase, where participants complete subjective evaluations.

2 Methodology

2.1 Participants

Ten university students (8 female, aged 20 to 25) participated in the experiment. One participant (subject 999, female) underwent EEG monitoring, while the remaining nine (7 female) completed an identical behavioral-only version of the paradigm to provide increased statistical power for behavioral analyses. All participants completed the experiment individually and were compensated with the monetary outcome of a randomly selected trial from either the Low- or High-stake block.

2.2 Stimuli

Stimuli consisted of synthetic price trajectories generated by a Geometric Brownian Motion (GBM) [16] process with a single embedded jump to simulate sudden volatility that often occur in financial markets. Formally:

$$S_t = S_0 \cdot \exp\left(\sum_{i=1}^t X_i\right) \cdot (1 + d \cdot J)^{\mathbf{1}_{[t \geq k]}}$$

where:

$$X_i \sim \mathcal{N}(\mu, \sigma^2), \quad J \sim \mathcal{N}(\mu_J, \sigma_J^2), \quad d \in \{-1, +1\},$$

$$k \sim \mathcal{U}\{[0.1T], \dots, [0.9T]\}$$

Here S_0 denotes the initial price and T the number of discrete time steps per trial. Crucially, the drift and volatility are not fixed but are drawn jointly per trial, $(\mu, \sigma) \sim \hat{F}_{\mu, \sigma}$, where $\hat{F}_{\mu, \sigma}$ is the empirical joint distribution estimated from non-overlapping 24-hour windows of eight years of Bitcoin price data; joint sampling preserves the natural correlation between drift and volatility regimes. All other draws are independent [6].

GBM was chosen as the base process because its log-normal returns approximate the statistical behavior of real financial assets while remaining fully parameterizable. As confirmed by a forced-choice discrimination task, the base simulation yields trajectories perceptually indistinguishable from real Bitcoin price windows (see Appendix A).

The embedded jump captures sudden, large-magnitude price dislocations of the kind that motivate jump-diffusion modelling in the first place [13]¹, such as flash crashes and earnings-driven gaps. Its direction constitutes a within-subject factor, its magnitude is drawn from a fixed distribution calibrated to realistic shock sizes, and its timing is randomized within the central 80% of the trial to prevent temporal anticipation. The resulting process is referred to as a jump-diffusion model (JDM) throughout.

¹The present model differs from Merton's (1976) since jump arrivals follow a fixed-rate schedule (exactly one per trial) rather than following a Poisson distribution.

2.3 Experimental Procedure

The experiment employed a within-subject design structured across three sequential blocks, each defined by the level of monetary capital at stake: a Control block (DKK 1 nominal, no real capital), a Low-stake block (DKK 50 per trial), and a High-stake block (DKK 100 per trial).

Each block drew from a pre-generated master plan of 80 trials (20 per condition), shuffled with a block-specific seed to ensure reproducibility across participants. From this plan, a per-session subset was drawn: behavioral participants completed 6 trials per condition (24 per block) plus a stimulus-validation phase, while the single EEG participant completed 18 trials per condition (72 per block) to improve the statistical likelihood of observing P300 - and FRN waves but without the stimuli-validation phase. However, due to time constraints, the High-stake block was terminated early, yielding only 44 of 72 trials ($\approx 61\%$).

As illustrated in Figure 1, each trial began with a position disclosure of a random fictional stock ticker. Proceeding, the pre-determined JDM process was plotted on a dynamic y-axis at a rate of one data point every 0.025 seconds for a total of 288 data points (representing the 5-minute windows across a 24-hour period). To clarify the fictional values and attempt to trigger a stronger counterfactual response, labels were added to show both the participant’s current position and the alternative outcome. Participants could execute their action and switch positions at any point during the JDM process. This action was limited to a single trial response to minimize muscle-movement artifacts in the EEG data.

Afterwards, participants were asked quantify their emotional experience of the trial using the Self-Assessment Manikin (SAM), specifically rating arousal, regret, and valence on a scale from zero to nine (see Appendix B).

2.4 Conditions

Crossing the initial position (ASSET vs. CASH) with jump direction yields four *intended* conditions (Table 1): an invested participant faces a Gain or a Loss, while an uninvested participant faces a foregone gain (FOMO) or an avoided loss (Relief). As both factors are fixed by the trial plan, the intended conditions are balanced by construction.

	Positive jump	Negative jump
Invested (ASSET)	Gain	Loss
Not invested (CASH)	FOMO	Relief

Table 1: Intended conditions, defined by initial position $\times$ jump direction.

The permitted action can, however, change the position held when the jump occurs, and it is this *realized* position that fixes the outcome the participant actually experiences. The intended condition is therefore distinguished from the *realized condition*, defined by the position held at the jump. Acting before the jump flips that position and, with direction constant, moves the trial to the opposite row of Table 1: selling maps ‘Gain’ to ‘FOMO’ and ‘Loss’ to ‘Relief’, buying maps ‘FOMO’ to ‘Gain’ and ‘Relief’ to ‘Loss’. Acting after the jump, or not at all, leaves the realized condition equal to the intended one.

Realized conditions are thus *not* necessarily balanced, since their distribution depends on participants’ choices. Affective and electrophysiological analyses are accordingly grouped by realized condition, with the intended-to-realized transition retained as a behavioral moderator.

2.5 EEG Acquisition and Preprocessing

Continuous electroencephalography (EEG) was recorded from one participant (subject 999) using a 64-electrode BioSemi system, with electrodes positioned according to the international 10-20 layout. Data were digitized at 2048 Hz inside a Faraday cage to minimize external electromagnetic interference.

Experimental triggers were stored on the BioSemi Status channel and decoded with the events-finding routine in MNE-Python. Due to an error with a BLOCK_START trigger code, block and realized condition were taken from the per-block behavioral records. The two surviving block-end markers were used to segment the spike sequence into the three blocks, resulting in a verified one-to-one mapping between EEG and behavioral data.

Preprocessing in MNE-Python included a 50 Hz filter to suppress European powerline interference. Continuous data were then resampled from 2048 to 256 Hz jointly with the event array to preserve spike timing

To reconcile the competing filtering requirements of slow Event-Related Potential (ERP) components and of Independent Component Analysis (ICA), two filtered copies of the continuous data were prepared. The analysis copy was band-pass filtered from 0.1 to 40 Hz; the gentle 0.1 Hz high-pass preserves the slow P300 component, which a more aggressive high-pass would attenuate. A separate copy was band-pass filtered from 1 to 40 Hz for ICA training, since a 1 Hz high-pass yields a cleaner decomposition. The ICA solution estimated on the 1 Hz copy was subsequently applied to the 0.1 to 40 Hz analysis copy.

One channel (PO4) was excluded prior to ICA on the basis

of a logged loose connection, confirmed by a signal variance roughly sixty times that of the next-noisiest channel. Two further channels (C1, P2) showed transient high-variance episodes but no persistent fault; lying near the parietal P300 ROI, they were retained and their transients deferred to ICA and epoch rejection.

ICA was computed on the 1 Hz copy with a fixed random seed for reproducibility (62 components, one below rank). Components were inspected visually using their scalp topographies, time courses, and power spectra, together with their per-segment variance profiles. Two components (components 0 and 6) were identified as ocular (a blink component and a lateral eye-movement component, both with frontal topographies, low-frequency-dominated spectra, and continuously elevated variance) and were removed.

Following artifact rejection via ICA, the excluded bad channel (PO4) was mathematically reconstructed using spherical-spline interpolation, computing its signal based on the spatial voltage topography of the remaining intact electrodes.

Cleaned data were segmented into epochs spanning -200 to +800 ms relative to spike onset and baseline-corrected to the -200 to 0 ms pre-spike interval. Epochs with peak-to-peak amplitudes exceeding $150 \mu\text{V}$ were rejected to eliminate residual high-voltage artifacts that survived the ICA, which excluded 5 epochs. Because roughly half of all trials contained an action, and such key-press and decision-related activity contaminates a spike-locked average, trials in which an action fell within the epoch window were subsequently excluded. In total, 151 out of 188 epochs were retained (Gain: 37, Relief: 24, FOMO: 32, Loss: 58).

2.6 EEG Statistical Analysis

The EEG data were analyzed using a data-driven Spatiotemporal Cluster-Based Permutation Test (SCPT). This algorithm evaluates all 64 electrodes simultaneously over specified temporal search windows. The Feedback-Related Negativity (FRN) and P3a components were evaluated within the 200–350 ms post-spike window, the P3b within the 300–600 ms window, and the LPP within the 400–800 ms window. For each test, adjacent significant data points were grouped into spatiotemporal clusters, and a custom null distribution was constructed by shuffling the trial labels 1,000 times, mathematically controlling for the multiple comparisons problem.

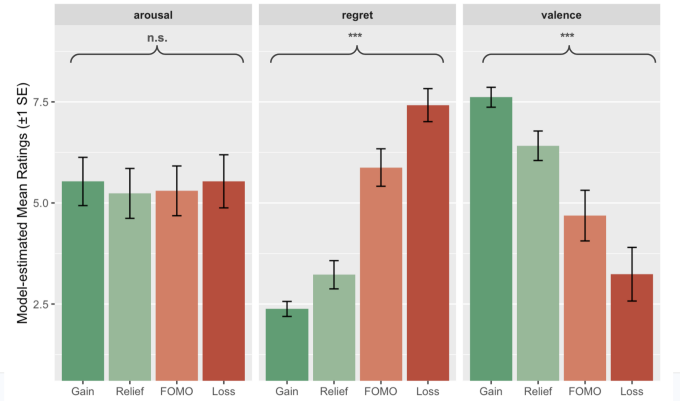


Figure 2: LMM-estimated population means of each emotion on the SAM rating scale.

3 Results

3.1 Behavioral Results

Unless explicitly stated otherwise, all subsequent behavioral analyses utilize Linear Mixed-effects Models (LMMs) specified with random intercepts and uncorrelated random slopes [10, 1] (see Appendix C). For all default mixed-model ANOVA tests, the Kenward-Roger approximation was used to determine denominator degrees of freedom, and all post-hoc pairwise comparisons were Tukey-adjusted.

SAM Ratings Across Conditions

In order to evaluate H2, the participants’ SAM ratings were analyzed. As suggested in Figure 2, the omnibus test revealed that while arousal ratings did not differ significantly across the four primary conditions ($F(3, 13.54) = 0.89, p = .469$), highly significant main effects of condition were detected for both subjective regret ($F(3, 14.48) = 60.79, p < .001$) and valence ($F(3, 13.78) = 24.55, p < .001$).

The follow-up pairwise comparisons (see Appendix D) revealed that the ‘Gain’ and ‘Relief’ conditions generated the lowest levels of subjective regret, with no significant difference observed between the two ($p = .128$). Conversely, the ‘FOMO’ and ‘Loss’ conditions generated the highest levels of regret ($p = .077$). All other pairwise comparisons were highly significant (all $ps \leq .001$) with moderate-to-huge effect sizes ($|d|s \geq 0.5$).

An inverse pattern emerged for valence. The ‘Gain’ condition was rated the most positive, whereas the ‘Loss’ condition was rated the least positive. Despite uniformly large-to-huge effect sizes across all conditions ($|d| > 0.88$), Relief-FOMO ($p = .085$) and FOMO-Loss ($p = .357$) were the only non-significant comparisons (all others $p \leq .015$).

Next, to examine H3, the four conditions were collapsed into a binary ‘Invested’ versus ‘Cash’ predictor. A linear mixed-

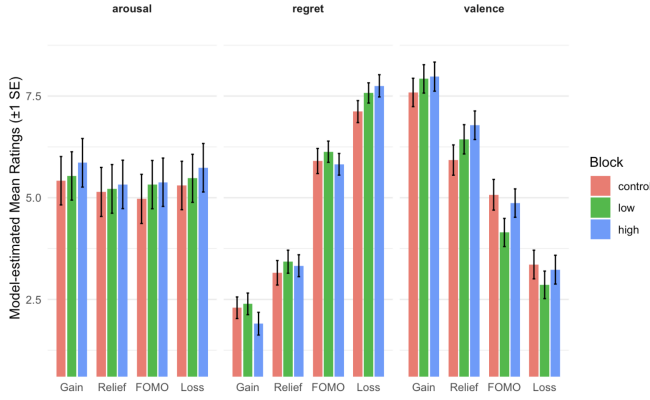


Figure 3: LMM estimated marginal means by block.

effects model revealed that this main effect of position was not significant ($F(1, 8.79) = 3.13, p = .11, d = 0.207$).

Blocks and SAM

To test H4, condition-by-block interactions were examined. As Figure 3 suggests, the influence of monetary stakes was minimal. Capital size did not significantly alter arousal ($F(6, 767.06) = 0.28, p = .948$) or regret ($F(6, 768.53) = 0.95, p = .461$). While a significant condition-by-block interaction emerged for valence ($F(6, 767.41) = 2.77, p = .011$), estimated marginal means revealed a small, U-shaped pattern. Thus, there was no clear evidence of linear emotional amplification driven by capital size.

Subsequent Risk Behavior

To evaluate H5, the real-stakes trials (excluding the control block) were isolated and fitted with mixed-effects logistic regressions predicting final position (Asset vs. Cash), controlling for randomized starting position. The behavioral data revealed a strong action bias, with participants executing a trade to switch positions in 81.35% of all trials. Consequently, starting position significantly predicted final behavior ($\chi^2(1) = 175.35, p < .001$) strictly by predicting its reversal. The predictor for preceding trial's financial outcome only approached significance ($\chi^2(3) = 6.67, p = .083$), indicating prior results did not systematically drive subsequent risk preferences.

3.2 EEG Results

FRN, P3a, and Effect of Agency

To evaluate H2 electrophysiologically, an SCPT was initially conducted comparing favorable (Gain and Relief) against unfavorable (Loss and FOMO) outcomes. This identified a significant fronto-central cluster ($p = .023$) spanning the 215–352 ms temporal window (see Figure 4).

To further investigate the drivers of this grouped effect, an exploratory 4-condition SCPT was conducted, revealing a highly significant main effect across the individual conditions ($p = .001$) within a similar timeframe (211–352 ms). Visual inspection of the individual grand-average waveforms revealed that the 'Gain' and 'FOMO' conditions (upward market spikes) exhibited a pronounced negative deflection, whereas the 'Loss' and 'Relief' conditions (downward market spikes) remained relatively flat.

Following this initial time window, a robust fronto-central positivity was observed uniformly across all four conditions, consistent with a P3a component (grand-average peak amplitude: $11.9 \mu V$ at 355.5 ms) as seen in Figure 4a.

To isolate the effect of agency (H3), the SCPT compared 'Active Loss' (buying into a negative spike) to 'Passive Loss' (already holding the asset). This, however, yielded no significant clusters ($p > 0.05$).

P3b and LPP Components

As illustrated in Figure 4b, a distinct centro-parietal positivity consistent with the P3b component was observed within the 300–600 ms temporal window (grand-average peak amplitude: $3.41 \mu V$ at 429.7 ms (sustained positivity rise accredited to LPP wave)). To evaluate whether the amplitude of this component scaled monotonically with capital size (H1), a SCPT was conducted comparing the Control, Low, and High stake blocks. The analysis identified no significant clusters ($p > .05$), indicating that P3b amplitudes did not significantly differentiate between the financial stake conditions.

To evaluate whether active capital exposure amplified late-stage neural processing (H3), the LPP was examined within the 400–800 ms temporal window. A SCPT comparing invested trials (the 'Gain' and 'Loss' conditions) against uninvested trials (the 'FOMO' and 'Relief' conditions) identified a highly significant positive cluster spanning ($p = .002$) persisting from 516 to 801 ms (see Figure 5c), demonstrating that an active market position elicited significantly larger positive amplitudes in this late temporal window compared to passive observation.

4 Discussion

This pilot study aimed to validate a continuous jump-diffusion paradigm and investigate the behavioral and neural correlates of financial evaluation, counterfactual processing, and risk under volatility. Overall, the experiment yielded a nuanced landscape of robust methodological validations alongside several theoretically significant null results.

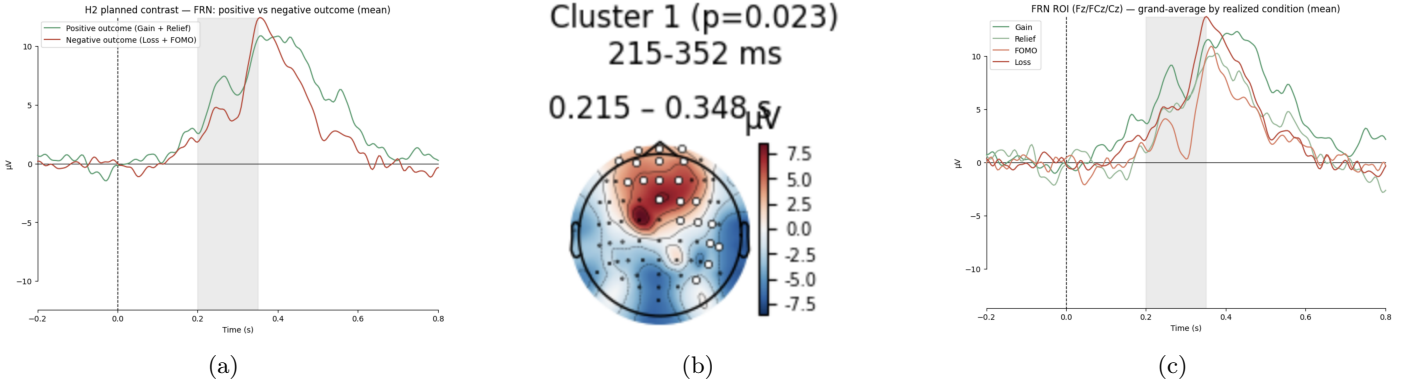


Figure 4: (a) Illustrates the divergence between outcome groups. (b) displays the scalp topography of the significant cluster, confirming the fronto-central focus typical of FRN-related processes. (c) The FRN ROI (Fz, FCz, Cz) displays a clear fronto-central negativity in the 200–350 ms window, consistent with established literature on prediction error signaling.

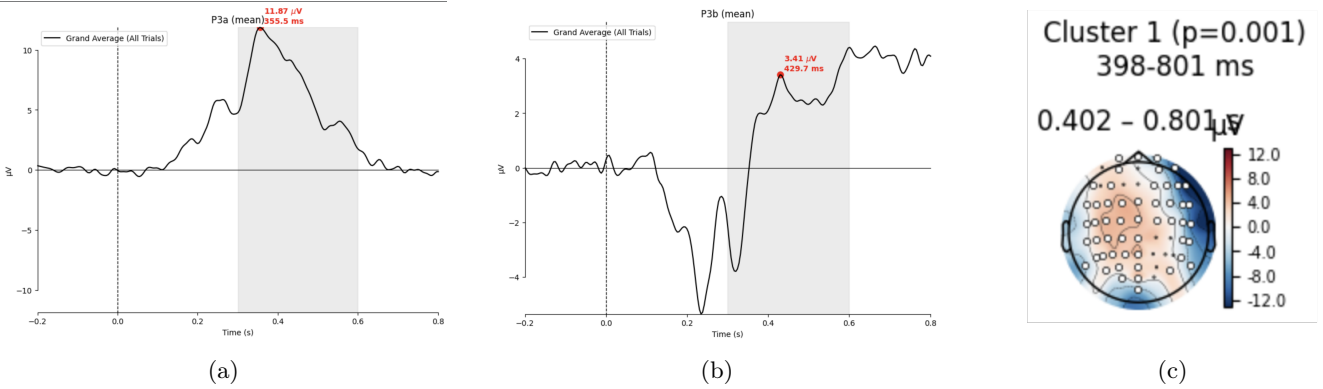


Figure 5: (a) P3a Grand Average (ROI: FZ, FCz, Cz) (b) P3b Grand Average (ROI: Pz, CPz, POz), (c) Scalp topography of LPP cluster.

4.1 Addressing the Hypotheses

P300 Elicitation and Ocular Masking

H1 posited that the continuous JDM would successfully elicit observable P300 and FRN components, validating its efficacy against established discrete-choice paradigms. The data only partially confirmed this fundamental methodological premise. Regarding early automatic components, H1 failed because a genuine cognitive FRN was not visible. The continuous presentation format introduced severe sensory confounds that completely masked the underlying FRN component under a wave of sudden tracking eye movements, as detailed extensively in the evaluation of H2.

However, the paradigm achieved clear validation regarding later cognitive and exogenous processing. A massive, distinct fronto-central positivity consistent with a P3a was clearly observable within the 300–400 ms post-spike window, followed by a classic centro-parietal P3b. Aligning with established literature on stimulus novelty and involuntary attention shifts [15], this confirms that a continuous, dynamically rendering market simulation can effectively evoke time-locked P300 complexes.

To validate these findings against traditional discrete architecture, publicly available EEG datasets for the Iowa Gambling Task (IGT) [3] and a Three-Armed Bandit Task (TABT) [4] were grand-averaged (see Appendix E). The IGT closely matches the JDM in P3a and P3b temporal alignment and shares similar parietal LPP activity (Pz). However, the static IGT signal is remarkably weak, yielding early voltages roughly 11% ($1.36\mu V$) of the massive P3a amplitudes generated by the continuous JDM. This vast discrepancy indicates that the sudden, real-time trajectory tracking inherent to the JDM commands far greater involuntary attentional resource allocation than static card-selection screens.

Conversely, comparisons with the TABT expose severe morphological divergences. While the TABT's early P3a temporal elements and voltage magnitudes ($8.46\mu V$) approach the JDM's initial novelty response, its subsequent P3b component is entirely inverted, presenting as a negative deflection. This complete divergence indicates that the late-stage cognitive evaluations performed during real-time asset crashes or surges share entirely different neural configurations than the passive reward-matching calculations of a discrete bandit task.

Thus, while the continuous presentation format obscures early prediction error signals, the JDM demonstrates a clear capacity to evoke a full, sequential P300 complex. It drives both robust exogenous attention shifts and subsequent endogenous parietal evaluations, though the latter appears to share distinct neural profiles depending on the task.

Realized vs. Counterfactual Outcomes

Hypothesis 2 (H2) posited that unfavorable financial outcomes (Loss and FOMO) would trigger lower valence, heightened regret, and a more pronounced FRN compared to favorable outcomes (Gain and Relief). Behaviorally, the LMMs revealed condition-driven fluctuations in subjective regret and valence. Specifically, the pairwise comparisons revealed no significant differences in either subjective regret or valence when comparing realized financial consequences to their counterfactual equivalents (i.e., Loss versus FOMO, and Gain versus Relief). This statistical equivalence suggests that within a continuous trading environment, the emotional impact of financial volatility is dictated just as heavily by simulated alternative scenarios as it is by tangible changes in capital.

Coupling the behavioral results with the electrophysiological data revealed an unexpected dissociation between early neural processing and subjective evaluation. While the initial SCPT misleadingly identified a significant fronto-central cluster between outcome groups that appeared to be an FRN component, the exploratory four-condition SCPT unmasked a different reality. Visual inspection suggested that this neural effect was driven by the physical direction of the jump rather than a reward prediction error (RPE), since Gain and FOMO share identical upward trajectories but zero psychological valence behaviorally.

This frontal asymmetry likely stems from sudden eye movements triggered by rapid market spikes. Because spike onset varies randomly (10%-90% of the trial), these non-systematic tracking movements slip through standard ICA filtering. These sudden upward glances leak directly into fronto-central channels (Fz/FCz/Cz) via dipole conduction, manifesting as the observed negative deflection. However, this explanation alone cannot account for the complete flatness of the downward conditions (Loss and Relief); if the artifact were purely a baseline visual response to any sharp trajectory change, sudden market crashes of equivalent magnitude should theoretically elicit a symmetric deflection.

Ultimately, H2 is only partially supported. While the behavioral findings coincide with the expected psychological consequences of realized and counterfactual financial outcomes, the electrophysiological aspect of H2 is firmly rejected. The continuous nature of the JDM introduces a

massive visual confound that could not be filtered out, completely masking the underlying cognitive FRN component.

Active Capital and Affective Engagement

H3 proposed that active capital exposure (i.e., "skin in the game") would increase subjective arousal and LPP amplitudes. The SCPT revealed a distinct positive cluster in the late temporal window for invested positions (Gain and Loss) compared to passive bystander positions (FOMO and Relief). While limited to a descriptive pilot profile, this divergence supports the behavioral predictions of the endowment effect. Neurally, it reflects the LPP's established role as an index of sustained affective engagement [5], suggesting a baseline tracking of active financial risk.

This neural trend mirrors the behavioral data, where the main effect of market position on self-reported arousal was non-significant but closely approached conventional significance. Although a divergence between explicit ratings and the single-subject trace could theoretically point to an unconscious process, the shared descriptive trend indicates that the lack of statistical significance reflects limited statistical power rather than a true psychological null.

Cognitive Ceilings and Task Habituation

Hypothesis 4 predicted that larger financial stakes would monotonically increase parietal P3b amplitudes and linearly amplify subjective emotional ratings. However, neither the neural nor behavioral data supported this. While a distinct P3b component emerged (300–600 ms), spatiotemporal cluster permutation tests revealed that amplitudes did not significantly differ across the Control, Low, and High blocks. Behaviorally, increased stakes failed to alter physiological arousal or subjective regret, producing only minor, non-monotonic fluctuations in valence.

These null findings likely stem from three interconnected methodological constraints. First, behaviorally, the lack of emotional scaling reflects a classic "lab effect" [8, 11]; because the funds were experimenter-provided and the 50 DKK increments were modest, the task may have failed to threaten personal wealth and lacked the emotional salience required to shift subjective ratings. Second, neurally, the insensitivity of the P3b component suggests a cognitive ceiling effect. The continuous, unpredictable visual monitoring inherent to the jump-diffusion paradigm may have saturated centro-parietal processing capacity, leaving limited attentional resources available for the capital blocks to further modulate P3b amplitudes. Finally, both of these limitations were exacerbated by the fixed, sequential presentation of the blocks. By the time participants reached the High-stake condition, they had already processed up to 144 volatile

jumps, leading to a progressive task habituation that neutralized any novelty-driven emotional or electrophysiological amplification intended by the monetary escalation.

Action Bias Over Risk Adjustment

Hypothesis 5 predicted that prior financial outcomes would dictate subsequent risk preferences, but the data rejected this. Prior outcomes had no significant effect; instead, participants exhibited a massive action bias, executing a trade to reverse their starting position in over 81% of all trials.

Rather than dynamically adjusting risk exposure based on recent performance, this constant trading likely reflects low-stakes gambling behavior. Because the funds were provided, participants had nothing to lose. Furthermore, since only one trade was allowed per trial, participants likely treated the task as an abstract gamble—swapping positions simply because they could, rather than executing a calculated financial strategy.

4.2 Limitations and Ecological Validity

The primary limitation of this pilot study is the constrained sample size ($N = 1$ EEG, $N = 10$ behavior), restricting neural findings to descriptive pilot data and leaving behavioral analyses underpowered for detecting subtle physiological shifts. Additionally, the high-stake block was prematurely terminated (44 instead of 72 trials), and allowing free trading choice created uneven trial distributions across conditions, further reducing statistical power and introducing noise into the averaged ERPs.

Regarding ecological validity, the jump-diffusion model demonstrated high visual and statistical realism. A discrimination task confirmed participants could not distinguish the synthetic trajectories from historical Bitcoin data ($BF_{01} = 5.40$). However, the mechanism of interaction lacked real-world equivalence. Executing a trade via a simple laboratory keystroke failed to evoke the psychological weight, ego-investment, and personal responsibility inherent to actual financial agency. Consequently, while the paradigm successfully emulated unpredictable market volatility, it could not fully replicate the deep emotional stakes necessary to neurologically amplify early prediction errors.

4.3 Future Studies

Future research must address the structural limitations of this pilot to better capture the neural correlates of financial decision-making. First, to overcome habituation and cognitive ceiling effects, subsequent designs must randomize or counterbalance the presentation order of the blocks. Second, achieving a fully powered sample size for both be-

havioral and EEG cohorts is essential; this will validate effect sizes and naturally resolve the uneven trial distributions caused by free-choice trading, preserving the ecological validity of participant agency. Third, concurrent eye-tracking should be integrated to actively regress the continuous visual tracking artifacts that masked the early FRN component. Finally, to mitigate the *lab money* effect and establish genuine psychological risk, future paradigms should endow participants with a substantial, highly apparent upfront capital pool (e.g., 1,000 DKK). Allowing each subsequent trade to directly compound or deplete this specific pool will better leverage the endowment effect and simulate authentic financial stakes.

5 Conclusion

This pilot study explored the gap between discrete neuroeconomic paradigms and the continuous, volatile nature of real-world financial markets. Using a Jump-Diffusion model based on historical data, we simulated the dynamic environment of retail trading to provide a preliminary framework for examining decision-making, counterfactual regret, and neural processing under continuous risk.

Behaviorally, continuous market simulations successfully induced complex psychological evaluations. The subjective experience of counterfactual outcomes closely mirrored realized consequences, with the regret of missing a surge rivaling the distress of an active crash. However, the anticipated scaling with increased financial stakes did not materialize. Likely influenced by an action bias and a lab money effect, participants treated the task as a risk-free gamble. This highlights the need for genuine psychological stakes, like a depletable upfront endowment, in future experimental designs.

Electrophysiologically, the continuous format yielded mixed results. The paradigm elicited late-stage cognitive components, specifically the P300 complex and the LPP, suggesting that active financial exposure commands involuntary attention and affective engagement. Conversely, the continuous visual tracking introduced severe ocular confounds that masked early reward prediction error signals, preventing the isolation of the Feedback-Related Negativity (FRN) component.

Ultimately, this study offers an early methodological step toward continuous neuroeconomic modeling while clearly defining its limitations. By illustrating what a synthetic paradigm might capture and noting where refinement is needed, such as concurrent eye-tracking and structural stakes adjustments, this research provides a foundation for future ecologically valid investigations into financial behavior.

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Appendix A Stimuli Validation

To assess the ecological validity of the base geometric Brownian motion (GBM) process, naive behavioral participants ($n = 9$) completed a real-versus-simulated discrimination task immediately after the main blocks. On each of 20 trials, a single price chart was animated to completion, after which the participant judged whether it had been drawn from real historical data or produced by the simulator (key [1] = real, [0] = fake). Each participant saw 10 real charts, reconstructed from the log returns of randomly drawn 24-hour Bitcoin windows (288 five-minute bars), and 10 simulated charts generated by the GBM process with per-window-sampled drift and volatility, presented in randomized order. All charts began from an identical nominal value of 100 DKK and shared identical axis scaling and animation timing, so that only the underlying price dynamics could inform the judgment. Across participants this yielded 180 classification trials (90 real, 90 simulated).

If the simulated paths were indistinguishable from real ones, discrimination accuracy should not differ from the 50% chance level. Participants classified 89 of 180 charts correctly (49.44%), marginally below chance. An exact binomial test did not reject the null hypothesis of chance-level performance ($p = .941$, 95% confidence interval [0.419, 0.570]).

Because a non-significant frequentist test cannot by itself provide evidence *for* the null, it was supplemented with a Bayesian estimate of the accuracy parameter θ . Placing a uniform Beta(1, 1) prior on θ and updating with the observed counts yields a Beta(90, 92) posterior, with mean 0.494 and 95% credible interval [0.422, 0.567]. This interval comfortably contains 0.5, and the posterior probability of above-chance performance is only 0.441. A Bayes factor (BF) for the same comparison favours the chance model: the data are approximately 5.40 times more likely under the hypothesis that $\theta = 0.5$ than under the alternative ($\text{BF}_{01} = 5.40$), constituting moderate evidence for chance-level discrimination on the Jeffreys scale.

Appendix B SAM Survey Prompts

Emotion	Prompt
valence	On a scale from 1 to 9, how are you feeling about the trial? Press a key from 1-9, where 1 = very negative and 9 = very positive
arousal	On a scale from 1 to 9, how emotionally intense was the trial? Press a key from 1-9, where 1 = very calm and 9 = very excited
regret	On a scale from 1 to 9, how much do you regret your decision? Press a key from 1-9, where 1 = no regret and 9 = extreme regret

Appendix C Model Comparisons

Table 2: Model Comparison Metrics Across SAM Ratings

Dimension	Model	Df	logLik	AIC	BIC
arousal	Random intercept	6	−1348.272	2708.544	2736.560
	Random slopes (uncorr)	9	−1327.409	2672.818	2714.843
	Random slopes (corr)	15	−1325.329	2680.659	2750.701
regret	Random intercept	6	−1606.876	3225.752	3253.769
	Random slopes (uncorr)	9	−1549.940	3117.879	3159.905
	Random slopes (corr)	15	−1540.399	3110.798	3180.840
valence	Random intercept	6	−1546.818	3105.637	3133.654
	Random slopes (uncorr)	9	−1419.465	2856.929	2898.955
	Random slopes (corr)	15	−1407.125	2844.249	2914.292

Note. Df = Degrees of Freedom; logLik = Log-Likelihood; AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion.

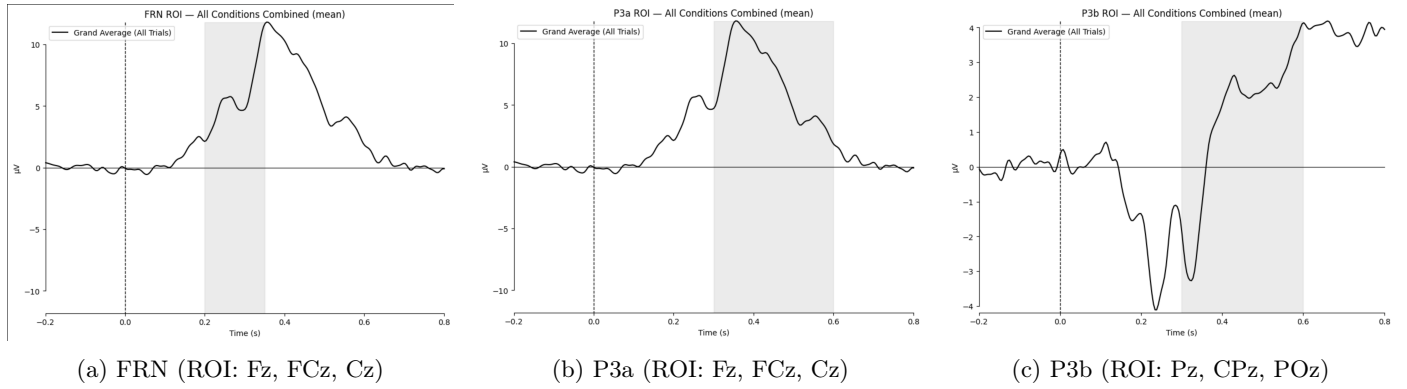
Appendix D Pairwise Comparisons for Regret and Valence

Table 3: Tukey-adjusted pairwise comparisons for regret and valence ratings.

Dimension	Contrast	ΔM	SE	p	d
Regret	Gain – Relief	−0.85	0.35	.128	−0.51
	Gain – FOMO	−3.50	0.46	< .001	−2.12
	Gain – Loss	−5.04	0.41	< .001	−3.06
	Relief – FOMO	−2.65	0.55	.001	−1.61
	Relief – Loss	−4.20	0.50	< .001	−2.55
	FOMO – Loss	−1.54	0.59	.077	−0.94
Valence	Gain – Relief	1.20	0.31	.015	0.88
	Gain – FOMO	2.93	0.60	.004	2.14
	Gain – Loss	4.38	0.63	< .001	3.20
	Relief – FOMO	1.73	0.65	.085	1.26
	Relief – Loss	3.18	0.69	.003	2.32
	FOMO – Loss	1.45	0.86	.357	1.06

Appendix E Validation of Data - A comparison to established literature

The following plots are the P300 (P3a and P3b) and FRN waves found in our study:



Plotted ERPs from the IGT:

Combined Grand Average Feedback ERP (Wins & Losses Grouped, N = 55)

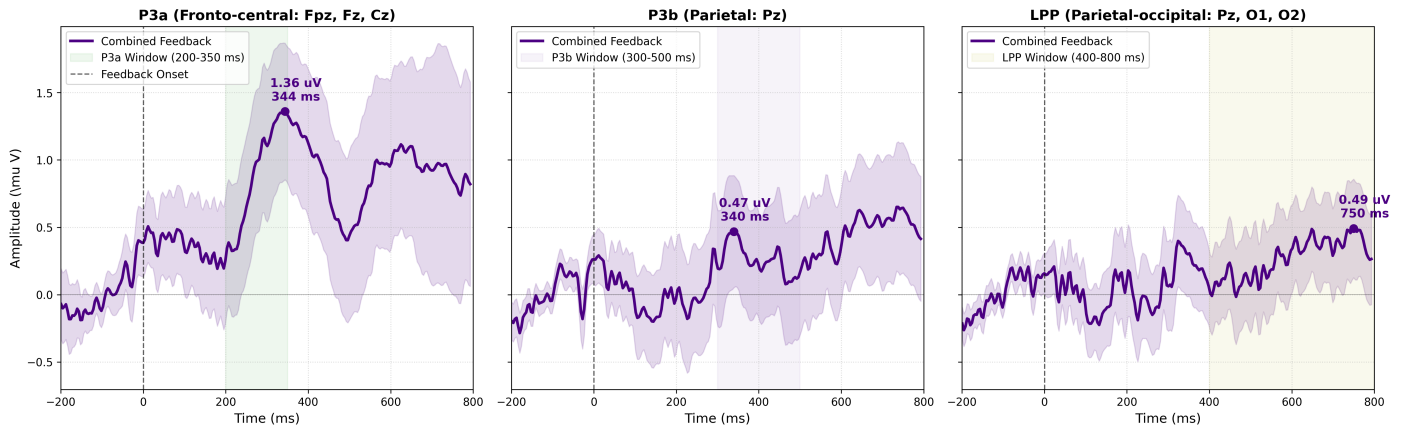


Figure 7: Iowa Gambling Task

Plotted ERPs from the TABT:

Combined Grand Average ERP Waveforms (Wins & Losses Grouped, N=23)

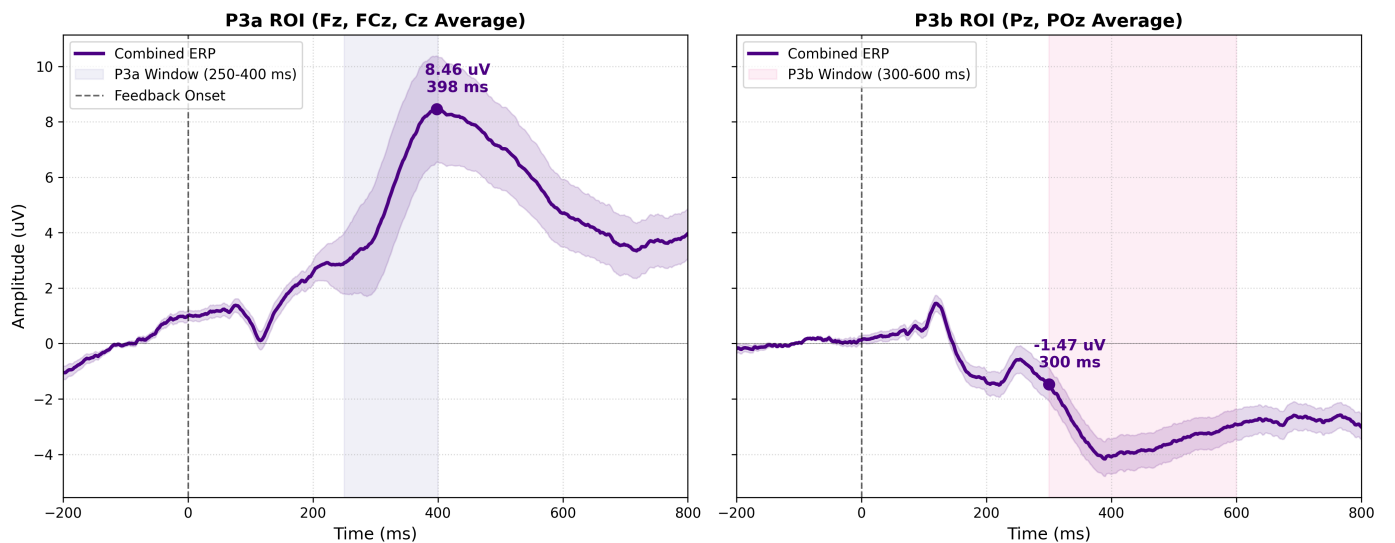


Figure 8: Three Armed Bandit Task